# Correct

## Streamlined Feedback and Autonomous Correction

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## INTRODUCTION

There are many automated writing aids for the native speaker that help correct spelling, grammar and style. These tools help native speakers by correcting and offering advice on possible mistakes and errors in their writing. These type of services would seem ideal as an aid to English language learners. Unfortunately they suffer from two problems. The first is that the interlanguage used in the correction and feedback of such programs is above the comprehension of the average learner often to the point of being useless or even detrimental to the learner. Secondly, these writing aids do not pick up and identify the types of mistakes that learners make. This is because the type of mistakes that learners make are very unlikely to ever be made by a native speaker and are thus not part of these programs . It is our hope that this program will be able to help non-native Japanese learners of English by identifying common errors that Japanese learners make and then offering live appropriate feedback to them so that they can achieve some level of autonomy in their writing while also allowing a teacher the time to concentrate on the more individual needs of the learner. It also frees the teacher to work on ideas like style, structure and form which are elements of writings that a teacher's feedback is essential for.

## Background

For improved writing, feedback is considered an essential element for both its development of student writing skills but also for increased motivation (Ferris, 2014; K. Hyland & Hyland, 2006). Offering students feedback that fits the students needs, is the appropriate level for the students to interpret and takes into account individual traits like aptitude or anxiety is very difficult (Carless, Salter, Yang, & Lam, 2011; Sheen, 2011). A teacher must be prepared to take into account the differences in learners' learning styles to offer feedback that allows active engagement (Katayama, 2007).

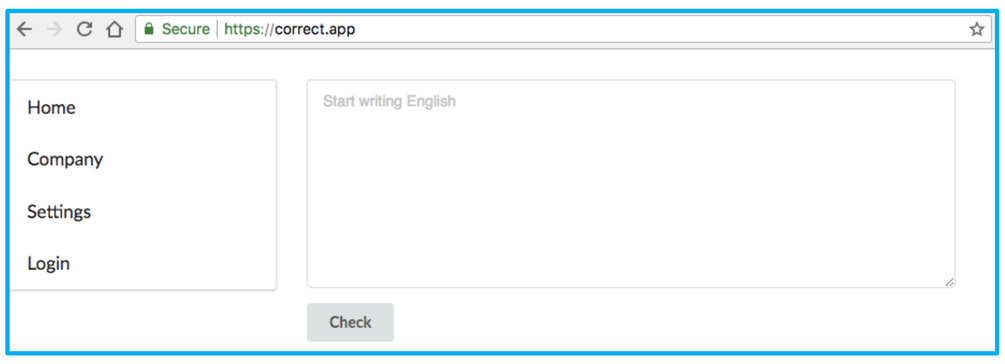
Because of the time constraints put upon the teacher in many classrooms some educators have tried alternatives to the feedback process such as peer review and technological assistance. This in combination with personal teacher counselling has proven to be highly effective. Therefore the time freed up by automating part of the feedback process should allow the teacher more time to offer complementary, bespoke feedback to individual students that combines to create improved outcomes for the learners.

It has been suggested that students are prone to make errors that fall into categorizable subsets (over-extention, fossilization etc.) that explain why the error is made (Touchie, 1986). It has also been noted that because errors share a common background causes, that students of English who share the same native language tend to make the same errors (Butler-Tanaka, 2000). Our work has shown that these repeated errors form a majority of the errors made by learners. Therefore, if a computer can identify and help a student correct these errors it will not only help the student to autonomously correct and stop making them but also free the teacher from repeatedly having to correct the same mistakes.

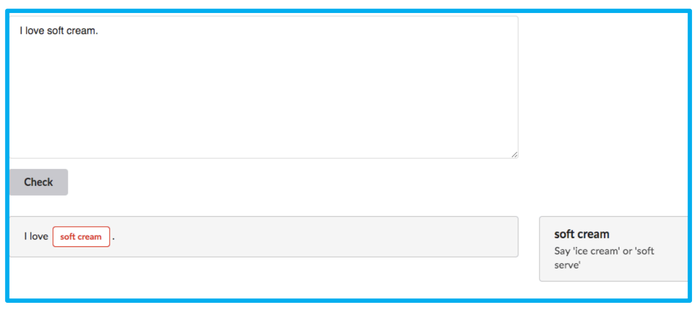
The errors identified for inclusion in the program come from data driven corpus research, previously published materials on Japanese learners mistakes and the combined experience of the two writers.

## How it Works

The UI was made deliberately simple and user friendly to not add any additional work to the writing process for students to use it effectively. The website consists of one page into which text can either be typed or copied and pasted.




Once the student hits the 'check' button. A list of errors and possible errors will be listened to the right of the text. These errors will come with advice (with the option of it being in either English or Japanese) for corrections and explanations of why this is or may be an error. We decided not to correct in real time as it slowed down or stopped student writing when they did not understand how to correct their mistake.



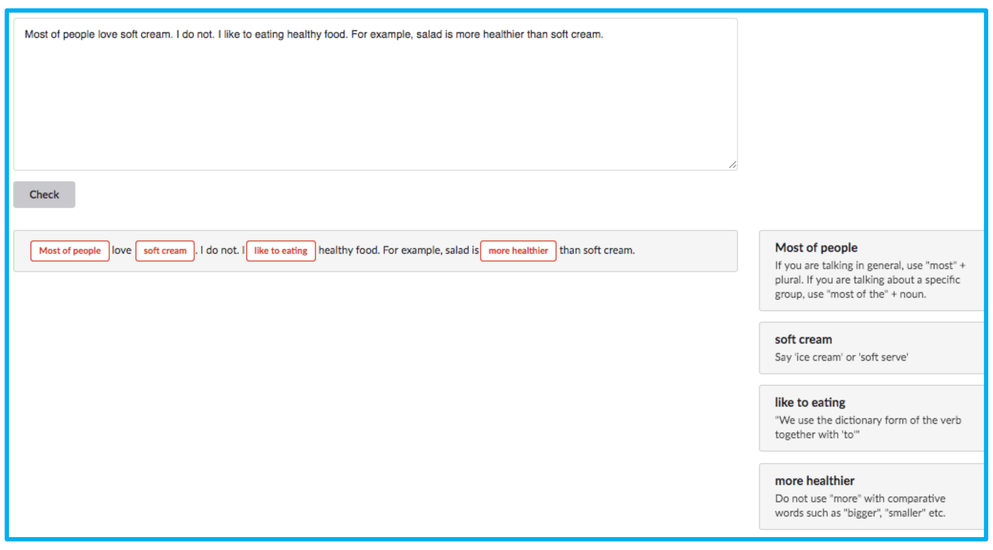

## Error Correction

The error correction offered by this program deals mostly with errors made at the local level (problems with individual words) but it also deals with some common but simple global mistakes at the sentence level. Global errors are identified by parsing the sentences and categorizing individual words into parts of speech.

| | | |
|---|---|---|
| idols? | Idol | Pop Star is the correct way to say idol. |
| anketo | enquête | Enquête is French. In English please use surv... |
| apato | Apartment | Apartment is the correct way to say this in A... |
| (arubaito\|bai... | A Part-time ... | Arubaito or Baito are German. In English say ... |
| (zubon\|jubon) | zubon | "Zubon" is wasei eigo. Say "pants" or "trousers" |
| zairu | zairu | "Zairu" is Japanese English. Use "rope" instead. |

| Pattern | Heading |
|---|---|
| one of the #Superlative #Singular | one the biggest things |
| one of #Superlative | one of the oldest |
| almost #Possessive? #Plural | almost + noun |
| like to #Gerund | like to (verb)-ing |
| more #Comparative | More + comparative |
| most #Superlative | most + superlative |
| one of the most #Adjective #Singular | one of the most + adjective + (singular noun) |

Errors have been categorised into different types so that learners will be able to understand the severity of the error. The different levels also indicate to some extent if there is a problem at all and if they may need to do further research or seek a teachers advice.




## Future Plans

The program aims to expand the amount of errors it can detect and offer advice on. The aim being to be able to correct a substantial number of the most common errors Japanese learners make.

As this tool is developed we intend to add features that will allow teachers to actively monitor students as use the app. This will allow the teacher to observe a class of students working on an assignment in real time.

It is also our intention to allow people to track their own mistakes over time so that they may see what of mistakes they make most.

We are keen to hear any feedback you may have today.

References:

Butler-Tanaka, P. (2000). Fossilization: a chronic condition or is consciousness-raising the cure? (Unpublished masters thesis). University of Birmingham, Birmingham, UK. Retrieved from https://www.birmingham.ac.uk/Documents/college-artslaw/cels/essays/matefltesldissertations/ButlerTdiss.pdf
Carless, D., Salter, D., Yang, M., & Lam, J. (2011). Developing sustainable feedback practices. *Studies in Higher Education, 36*(4), 395–407. https://dx.doi.org/10.1080/03075071003642449
Ferris, D. R. (2014). Responding to student writing: Teachers' philosophies and practices. *Assessing Writing, 19*, 6–23. https://doi.org/10.1016/j.asw.2013.09.004
Hyland, K., & Hyland, F. (2006). Feedback on second language students' writing. *Language Teaching, 39*(2), 83. https://dx.doi.org/10.1017/S0261444806003399
Katayama, A. (2007). Japanese EFL students' preferences toward correction of classroom oral errors. *Asian EFL Journal, 9*(4). Retrieved from http://www.asian-efl-journal.com/Dec_2007_ak.php
Sheen, Y. (2011). Corrective feedback, individual differences and second language learning. New York: Springer.
Touchie, H. (1986). Second language learning errors their types, causes, and treatment. JALT Journal, 8 (1), 75-80.

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